

Q) Print:-

```
*
* *
* * *
* * X *
```

```
#include <iostream>
using namespace std;
```

```
int main () {
```

```
    int n;
```

```
    cin >> n;
```

```
    for (int i=1 ; i<=n; i++) {
```

```
        for (int j=1 ; j<=i; j++) {
```

```
            cout << "X ";
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
    return 0;
```

```
}
```

Q) Print:-

```
a b c d
```

```
a b c d
```

```
a b c d
```

```
a b c d
```

```
#include <iostream>
```

```
using namespace std;
```

```
int main () {
```

```
    int n;
```

```
    cin >> n;
```

```
if (n < 1 || n > 26) {
```

```
    cout << "Invalid Input ! Enter between 1-26" << endl;
```

```
    return 0;
```

```
}
```

```
else {
```

```
    for (int i=1 ; i<=n; i++) {
```

```
        for (int j=97 ; j<=97+n; j++) {
```

```
            cout << (char)j << " ";
```

```
        }
```

```
        cout << endl;
```

```

    }
    return 0;
}

```

Q) Print:-

```

1
1 2
1 2 3
1 2 3 4

```

```

for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= i; j++) {
        cout << j << " ";
    }
    cout << endl;
}

```

```

    }
    return 0;
}

```

```

#include <iostream>
using namespace std;

```

```

int main() {
    int n;
    cin >> n;
}

```

Q) Print:-

```

a a a a
b b b b
c c c c
d d d d

```

```

if (n < 1 || n > 26) {
    cout << "Invalid" << endl;
    return 0;
}

```

else {

```

    for (int i = 97; i <= 97 + n; i++) {
        for (int j = 1; j <= n; j++) {
            cout << (char)i << " ";
        }
        cout << endl;
    }
}

```

```

    }
    return 0;
}

```

```

#include <iostream>
using namespace std;

```

```

int main() {
    int n;
    cin >> n;
}

```

A Second approach can be:

```
for(char ch='a' ; ch < 'a'+n; ch++) {
    for(int j=1; j <= n; j++) {
        cout << ch << " ";
    }
    cout << endl;
}
```

Q) Print:-

```
1
2 2
3 3 3
4 4 4 4
```

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cin >> n;
```

```
for(int i=1; i <= n; i++) {
    for(int j=1; j <= i; j++) {
        cout << i << " ";
    }
    cout << endl;
}
return 0;
```

Q) Print:-

```
1
2 1
3 2 1
4 3 2 1
```

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cin >> n;
```

```
for(int i=1; i <= n; i++) {
    for(int j=i; j >= 1; j--) {
        cout << j << " ";
    }
    cout << endl;
}
return 0;
```

Q) Print:-

```
a
b b
c c c
d d d d
```

```
#include <iostream>
using namespace std;
```

```
int main() {
    int n;
    cin >> n;
```

```
char ch = 'a';
```

```
for (int i = 1; i <= n; i++) {
```

```
    for (int j = 1; j <= i; j++) {
```

```
        cout << ch << " ";
```

```
    }
```

```
    ch++;
```

```
    cout << endl;
```

```
}
```

```
return 0;
```

```
}
```

A Second approach can be

```
char ch = 'a';
```

```
for (int i = 1; i <= n; i++) {
```

```
    for (int j = 1; j <= i; j++) {
```

```
        cout << c + (i - 1) << " ";
```

```
    }
```

```
    cout << endl;
```

```
}
```

Q) Print:-

```
* * * *
* * *
* *
*
```

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int n;
```

```
    cin >> n;
```

```
    for (int i = n; i >= 1; i--) {
```

```
        for (int j = 1; j <= i; j++) {
```

```
            cout << " * ";
```

```
        }
```



```

    cout << endl;
}
return 0;
}

```

A Second approach can be:-

```

for (int i=1; i<=n; i++) {
    for (int j=1; j<= n-i+1; j++) {
        cout << "*" << " ";
    }
    cout << endl;
}

```

Q) Print:-

```

1 2 3 4
1 2 3
1 2
1

```

```

#include <iostream>
using namespace std;
int main() {
    int n;
    cin >> n;
}

```

```

for (int i=1; i<=n; i++) {
    for (int j=1; j<= n-i+1; j++) {
        cout << j << " ";
    }
    cout << endl;
}
return 0;
}

```

A Second approach can be:-

```

for (int i=n; i>=1; i--) {
    for (int j=1; j<= i; j++) {
        cout << j << " ";
    }
    cout << endl;
}

```

Q> Print :-

```
4
4 3
4 3 2
4 3 2 1
```

```
#include <iostream>
using namespace std;
```

```
int main () {
    int n;
    cin >> n;
```

```
for (int i = 1 ; i <= n ; i++) {
```

```
    for (int j = n ; j >= n - i + 1 ; j--) {
```

```
        cout << j << " ";
```

```
    }
    cout << endl;
```

```
    }
    return 0;
```

Q> Print the following pattern, if $n = 5$

```

      *
     **
    ***
   ****
  *****
```

Analysis →

Star	Row	Space
1	1	4
2	2	3
3	3	2
4	4	1
5	5	0

Logic →

(i) Row = 1 to Row ≤ 5

(ii) Print Space, (5 - row) times or (n - row) times

(iii) Print Star row times